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Effects of central and peripheral Go/No-Go signals on reactions to peripheral stimuli in virtual reality

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Introduction: Reactions and their inhibition are crucial in many sports. Visual reaction times (RTs) typically increase with eccentricity, and inhibition tasks produce slower RTs than simple reaction tasks. In sports, responses to peripheral stimuli often depend on an additional signal. This randomized within-subject study examined how the position of additional Go/No-Go-signals affects RTs to peripheral stimuli.
Method: Healthy adults (n = 30; eligibility: no red-green blindness, no motor restrictions) performed a virtual reality reaction task to stimuli at varying eccentricities in three conditions using a virtual reaction wall. In the no-signal condition, participants reacted to illuminating buttons. In the two Go/No-Go conditions, an additional signal (green = Go, red = No-Go) appeared either centrally (central-signal condition) or peripherally (45° horizontally left and right; peripheral-signal condition). The primary outcome was RT to Go stimuli. Condition order was randomized for each participant using a computer-generated allocation.
Results: Three participants were excluded due to high error rates. Data from the remaining 27 participants showed that RTs increased with eccentricity. Reactions were fastest in the no-signal condition, followed by the central-signal condition, and slowest in the peripheral-signal condition. No evidence for differences in the slope of the eccentricity-related RT increase was observed.
Conclusion: This virtual reality setup replicated known real-world findings: RTs increase with eccentricity and are slower in inhibition tasks. The position of a Go/No-Go-signal influences absolute RTs, but no evidence for faciliatory or limiting influence of attentional shifts induced by the additional signals was found.

KEYWORDS
head-mounted display, inhibition, peripheral vision, reaction times, sports, virtual reality, visual perception

1 Introduction

Virtual reality (VR) refers to computer-generated environments that users can perceive and interact with, similar to the real-world (RW) (Parsons et al., 2017). Thereby, VR offers new possibilities in research due to several advantages over the real world (RW), such as controllable conditions and the ability to create scenarios that are difficult to realize in RW (Düking et al., 2018). Often, the ecological validity of experimental results is enhanced, as VR allows researchers to conduct tests in field-like settings while simultaneously collecting data (Draschkow, 2022). Therefore, VR can be a valuable tool for testing abilities essential in sports, such as responsiveness (Pastel et al., 2025a) and decision-making (Pastel et al., 2025b), and cognitive training regarding concentration and executive functions (Lachowicz et al., 2025; Lachowicz et al., 2024).

Nevertheless, results obtained in VR must be interpreted in relation to the RW, as differences exist between the two settings. For example, VR can, in some cases, induce cybersickness, which may impair performance in reaction tests (Nalivaiko et al., 2015), and the vergence-accommodation conflict creates different viewing conditions in VR due to a fixed distance between eyes and display, potentially affecting perception and performance (Hoffman et al., 2008). Furthermore, the field of view (FOV) in VR is often reduced, and vision is distorted in the peripheral visual field, leading to larger head movements in vision tasks compared to the RW (Pfeil et al., 2018). In addition, the VR setup often introduces latencies, e.g., between a controller movement and the movement shown on the display, affecting the measurements of reaction times (RT) (Langer et al., 2023). Therefore, often longer RTs are measured in VR, however, correlations between RTs in VR and RW are observed (Langer et al., 2023; Polechoński and Langer, 2022; Pastel et al., 2025a), possibly enabling transferability of VR results into the RW.

VR is already used in many different kinds of sports (Witte et al., 2025), as increasing ecological validity for training and testing methods is of great interest (Parsons and Barnett, 2019; Voinescu et al., 2023). Fast and accurate responses are crucial in many sports (Klos et al., 2019). This includes not only selecting the best response option, but also inhibiting incorrect reactions (Logan and Cowan, 1984). The ability to quickly and appropriately process the given information and decide whether to initiate or withhold a response is often linked to athletic performance (Heilmann et al., 2022). In football, for example, a ball-carrying player must quickly decide whether to pass to an open teammate but must simultaneously evaluate whether the pass could be intercepted, requiring inhibition (Heilmann et al., 2022). Such abilities are commonly examined using Go/No-Go tasks (Raud et al., 2020). Studies have shown that in these tasks, professional athletes tend to outperform amateurs (Verburgh et al., 2014), which has also been demonstrated using VR (Imperiali et al., 2025).

In simple reaction tests, the stimulus is detected by sensory receptors and transmitted via sensory neurons to the cerebral cortex, where the motor response is initiated through motor neurons (Yunpeng et al., 2024). In tasks involving inhibition, like Go/No-Go tasks, the stimuli must first be evaluated, leading to increased reaction times (RTs) (Danek and Mordkoff, 2011). A prominent explanation for this difference is Donders' subtractive method (Donders, 1969), which states that additional cognitive processes, like stimulus discrimination, prolong the RT. However, the subtractive method is considered outdated, as the motor processing time may also differ between the tasks, as participants prepare the movement but initiate an inhibition process that is released as soon as the go-signal is fully processed (Danek and Mordkoff, 2011). Regardless, RTs in simple reaction tasks are shorter than in Go/No-Go tasks (Squella and Ribeiro-do-Valle, 2003; Kida et al., 2005), which was also observed in VR (Imperiali et al., 2025; Loushy Kay et al., 2025), although not explicitly tested in that study.

In sport settings, information often appears throughout the entire visual field, indicating that stimuli (e.g., teammates to whom to pass) and additional signals (e.g., opponents possibly intercepting the pass) can occur centrally or peripherally. Thereby, peripheral vision is often used to monitor several objects while avoiding time-consuming saccades (Vater et al., 2022). The definition of what is considered centrally or peripherally differs widely. Throughout this paper, central perception is referred to <8° eccentricity (deviation from the fixation point, respectively the display center), near-peripheral to 8°–30° eccentricity, mid-peripheral to 30°–60° eccentricity, and far-peripheral to beyond 60° eccentricity as proposed by (Simpson, 2017). Several studies have examined the influence of stimulus eccentricity on RTs in simple reaction tests and tests similar to Go/No-Go tasks:

RTs to peripheral stimuli are generally longer than to central stimuli and increase with eccentricity (Strasburger et al., 2011). This is largely attributed to the distribution of photoreceptor and ganglion cells in the human retina (Curcio and Allen, 1990; Curcio et al., 1990). The number of ganglion cells (responsible for signal transmission to the brain) decreases in the visual periphery (Curcio et al., 1990), as does cone density (used for sharp and colored vision), whereas rod density (used for mesopic vision and movement detection) increases (Curcio and Allen, 1990). Furthermore, it has been shown that attention abilities decrease with stimulus eccentricity (Staugaard et al., 2016). Only a few studies have investigated this phenomenon in VR. Two studies examined reactions in basketball defense by presenting first-person-perspective 360° videos using either a Cave Automatic Virtual Environment (CAVE, a room where the virtual environment is projected to walls surrounding the user) (Vater, 2024) or a head-mounted display (HMD) (DeCouto et al., 2023). Both studies demonstrated that the RTs increase with eccentricity, particularly in the far-peripheral area (>60° eccentricity). Additionally, Bürger et al. (Bürger et al., 2024) reported longer RTs to peripheral stimuli in VR compared to RW but high correlations between both conditions. This finding is consistent with several studies comparing RTs to foveal stimuli between RW and VR (Langer et al., 2023; Pastel et al., 2025a). Regarding RTs in Go/No-Go tasks, also increases when Go- and No-Go-stimuli are briefly presented peripherally were observed (Sun and Varshney, 2018; Thorpe et al., 2001). This increase is nearly linear up to 70° eccentricity, and accuracy decreases substantially, dropping to around 60% at 70° eccentricity in Thorpe et al. (Thorpe et al., 2001). Both studies used large curved screens to display the stimuli at different eccentricities, which could be considered as semi-immersive VR in contrast to the fully-immersive HMDs or CAVE.

In the previously mentioned tasks, participants responded to a single stimulus. However, in sport contexts, multiple stimuli are perceived simultaneously, attracting attention (Klos et al., 2019) but having varying importance for the planning of actions. Not all of these stimuli may be task-relevant, but they may influence performance nevertheless. In this work, different stimuli are discriminated: stimulus (the main stimulus that should be reacted

Abbreviations: RT, Reaction Time; VR, Virtual Reality; RW, Real-World; FOV, Field of View; CAVE, Cave Automatic Virtual Environment; HMD, Head-Mounted Display; SSQ, Simulator Sickness Questionnaire; VA, Viewing Angle; LMM, Linear Mixed-Effects Model; VA_transf, z-transformed Viewing Angle; LnRT, Logarithmic Transformation of Reaction Time.

to, e.g., a free teammate), distractor (stimulus that distracts from the actual task, e.g., fans on the side of the pitch), cue (stimulus that signals the presentation of the main stimulus, e.g., a coach pointing toward the free teammate), and signal (stimulus that modifies the task, e.g., a possibly pass intercepting opponent). Several studies have examined the influence of task-irrelevant distractors or cues on the RT, with distractors, cues, and goal stimuli at various (mostly small) eccentricities. Central distractors impair RTs to central stimuli more than to peripheral stimuli (Beck and Lavie, 2005; Chen, 2008), whereas peripheral distractors have minimal influence on RTs to central stimuli (Beck and Lavie, 2005; Chen, 2008; Goolkasian, 1994) but significantly slow responses to peripheral stimuli (Chen, 2008; Chen and Treisman, 2008; Goolkasian, 1994). Although these distractors were task-irrelevant, such findings underscore the importance of spatial attention in reaction tasks involving multiple stimuli at different eccentricities. Furthermore, RTs can be shortened when additional task-irrelevant cues prime the stimulus location, as attention shifts towards the cued area, facilitating the processing of the stimulus (Facoetti, 2001). Similarly, at small eccentricities (8°), priming the stimulus in a Go/-No-Go task reduces the RT, which was the opposite for simple reaction tasks (Squella and Ribeiro-do-Valle, 2003).

Although many studies have examined the influence of stimulus eccentricity and task-irrelevant distractors and cues, no research has been found investigating the influence of task-relevant signals at different positions on the RT to goal stimuli at varying eccentricities. In many sports situations, the signal determining whether a player should react, and the stimulus to which they respond, are spatially separated. For example, in football, an opponent's pass (stimulus) might trigger a reaction depending on whether teammates or opponents are positioned to receive the ball (signal).

This study, therefore, aimed to investigate how the position of an additional task-relevant signal, which determines whether a response to a goal stimulus is required (i.e., converting the task into a Go/No-Go structure), influences RTs to stimuli at varying eccentricities in VR. To compare the RTs, the reaction test was conducted without an additional signal, with an additional signal in the central visual field, and with additional signals in the peripheral visual field. Based on previous work, we expected:

1. Longer RTs for stimuli at greater eccentricities, as shown previously (Strasburger et al., 2011)
2. Longer RTs with additional signals than without, due to the need to process the signal (Danek and Mordkoff, 2011)
3. Longer RTs with peripheral compared to central additional signals, due to slower processing in the periphery (Strasburger et al., 2011)

While these assumptions are well supported by the RW literature, an exploratory examination of an interaction between the position of the additional signal and the stimulus eccentricity is performed. We aim to analyze whether shifts in spatial attention, induced by the additional signal, facilitate or impair responses in the respective areas (Facoetti, 2001).

The study is thereby considered a sport-inspired perceptual-motor assessment, and aims to contribute to the development of VR-based test and potentially training tools for cognitive functions in sports.

2 Materials and methods

2.1 Participants

Thirty participants (9 females, 21 males, age: 25.8 ± 3.9 years) were recruited through self-selection and took part in the study after giving their written informed consent. A formal sample size calculation was not performed, as this was an exploratory study. A sample size of 30 was chosen based on feasibility and prior similar VR studies. Ethics approval was granted by the ethics committee of the first author's university. All participants were active in sports at a self-reported moderate to ambitious level (between training 2–3 times per week and daily). Except for five participants, all had prior VR experience, mainly through previous similar studies or gaming. Exclusion criteria involved motor impairments that would restrict the execution of the task, red-green color blindness, which was determined using Ishihara plates, and wearing glasses during the experiment (contact lenses were accepted), as the glasses frame could limit the FOV in the HMD.

2.2 Design

The study was designed as a within-subject trial with an exploratory framework and conducted in a single session lasting approximately 30 min. At the start, participants completed questionnaires concerning their demographic data, prior VR experiences, and their baseline cybersickness (the Simulator Sickness Questionnaire (SSQ) (Kennedy et al., 1993)).

Subsequently, three reaction tests were performed, each consisting of three 1-min rounds. The order of the tests was randomized for each participant and counterbalanced using a computer-generated random sequence in MATLAB (version R2022b). Randomization was simple, with equal allocation to each possible condition. The sequence was implemented by one of the experimenters. No allocation concealment was possible, as all conditions were present to each participant within a single session. Neither participants nor experimenters were blinded to condition assignment, as all conditions were apparent during task execution. One condition only involved reacting to randomly illuminating stimuli in different areas of the visual field (no-signal condition). The other two conditions were designed similarly to a Go/No-Go task and contained either an additional central signal (0° viewing angle, central-signal condition) or additional peripheral signals (45° to each side, peripheral-signal condition). These additional signals were presented as colored spheres indicating whether to respond (green = Go) or inhibit the response (red = No-Go) to an illuminating stimulus.

After the final reaction test, the SSQ was completed again to check for changes in cybersickness symptoms. Subsequently, a feedback questionnaire addressing the difficulty and potential problems of each test was completed.

The study design is illustrated in Figure 1.

2.3 Experimental apparatuses

The virtual environment was created using Blender (version 3.2.1), and the tests were programmed using customized C#-scripts within Unity (version 2021.2.8f1). The application was then

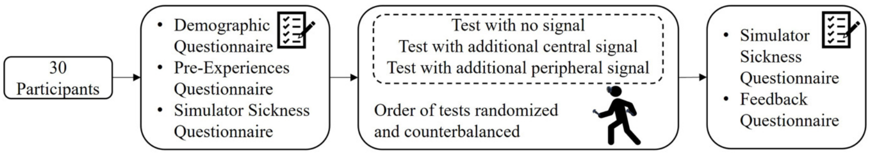

FIGURE 1
Visualization of the study design.

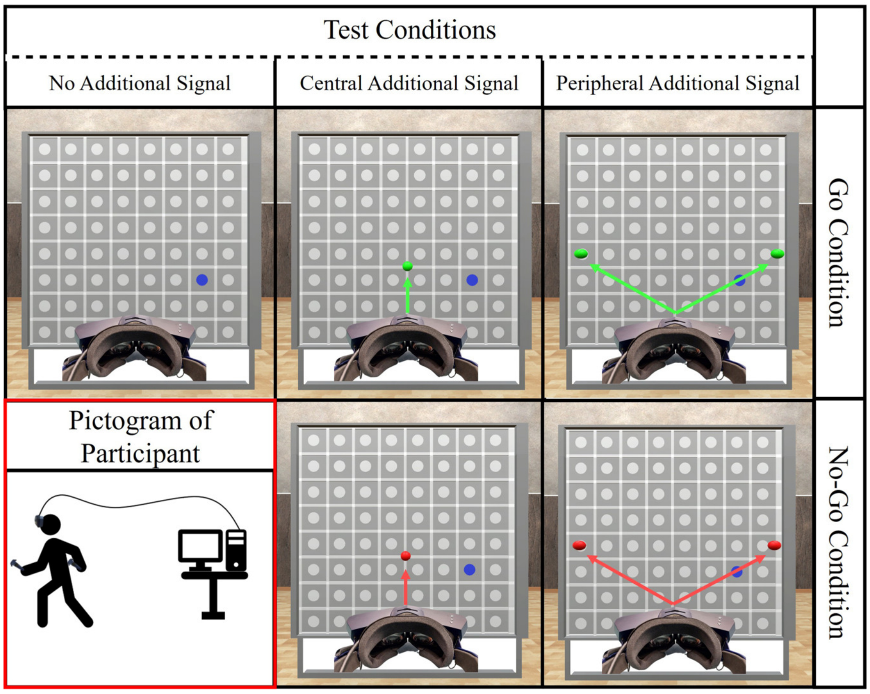

FIGURE 2
Visualization of the reaction wall in the three test conditions. A target lit up (blue) and in the central-signal and peripheral-signal conditions, an additional signal indicated whether to react (green, top row) or to inhibit the reaction (red, bottom row). The participants wore an HMD connected to a computer via cable (bottom, left).

exported to a standalone build and run with Steam (packet version 1722028579) with VR functionality enabled through SteamVR (version 2.5.4). Data were processed in MATLAB (version R2022b), and statistical analyses were conducted in R (version 4.4.3) with the packages lme4 (version 1.1.35.5), lmerTest (version 3.1.3), effectsize (version 1.0.0), and sjstats (version 0.19.1).

The VR setup included a Pimax Vision 5K Super HMD (manufacturer information: 200° diagonal FOV, 2,560 × 1,440 pixels per eye, 90 Hz refresh rate), connected via cable to a PC (Intel Core i7 CPU (11700F), 32 GB RAM, Nvidia RTX 4080). The HMD was used because it provides the greatest known FOV. Virtual hands were visualized in Unity using the position of two handheld HTC VIVE Controllers (2018) and two SteamVR Base Stations 2.0 tracked the HMD's and controllers' positions.

The virtual environment consisted of a simple room, with a virtual replica of the twall® Premium 64 (IMM electronics GmbH) (8 × 8 stimuli, each 10 cm in diameter, 22.5 cm between stimuli' centers) in its center. The real version of the twall®, although not being the gold standard in RT assessments, has been used in studies before (Müller et al., 2023; DeVocht et al., 2016). The height of the wall was individually adjusted depending on the participant's height, so that the top row of stimuli was 30 cm above the participant's head, an advantage that the real version cannot provide. The size of the reaction wall remained constant for each participant and was only vertically shifted. The setup for the three conditions is visualized in Figure 2.

2.4 Procedure

Since the task was simple, and in order to reduce cognitive load, no practice trials were completed. Instead, the first three Go-trials were considered familiarization and eliminated from the analysis. Each condition was conducted for three rounds lasting 1 min, with a one to 2-min break in between. Between the conditions, there was a break of approximately 5 min. Videos of the three conditions are presented in the Supplementary Material.

10.3389/frvir.2026.1706034

2.4.1 No-signal condition

Participants stood in front of the reaction wall, with all stimuli initially white. After verbally communicating their readiness, one of the 64 stimuli randomly lit up in blue and was to be virtually touched with either controller. Upon virtual contact, the controller briefly vibrated, and the stimulus turned white again. After 0.4 s, another random stimulus lit up. The time interval was chosen to encourage head movements, as participants started searching right after a stimulus was touched. Stimulus sequences were random and varied between conditions, rounds, and participants. This condition was completed without an additional Go/No-Go signal and served as a baseline condition.

2.4.2 Central-signal condition

The test principle remained the same but was extended by an additional Go/No-Go-signal. The signal consisted of a small sphere (5 cm in diameter) fixed in the center of the rendered image, moving with the participant's head but unaffected by eye movements, as sufficient eye tracking was unavailable due to the hardware model. The sphere was always either green (Go) or red (No-Go). For green signals, the task mirrored the no-signal condition. For red signals, participants were to inhibit responses. The stimuli in a No-Go condition turned white automatically after 1.5 s. The sphere's color was randomly determined and changed if needed, simultaneously with a stimulus being lit up. When the same condition (Go or No-Go) appeared consecutively, the sphere maintained the color uninterrupted. In 75% of the trials, the sphere was green (Go) and otherwise red (No-Go), a common distribution in Go/No-Go tasks (Raud et al., 2020). The order was random and not controlled between participants or rounds.

2.4.3 Peripheral-signal condition

This condition was identical to the central-signal condition, except that there were two spheres (signals), horizontally shifted 45° to the left and right of the center. Again, these spheres moved according to the orientation of the participant's head. Both spheres always showed the same color (green or red) and changed color, as in the central-signal condition.

2.5 Data processing

During all tests, 3D position data were recorded at 90 Hz for the stimulus, both controllers, the HMD, and an approximated gaze position (center of the HMD's FOV). For each stimulus, the hand used to react, the time point when the stimulus was first visible to the participant, and whether it was a Go- or No-Go trial were recorded.

No RTs were calculated for No-Go trials, since participants should not react in these trials. When participants responded during No-Go trials, this was marked as an error. For Go-trials and all trials in the no-signal condition, RTs were calculated from stimulus onset to the first movement toward the stimulus, using a custom MATLAB script.

Go-trials immediately following No-Go-trials were excluded, since RTs were much slower in these cases.

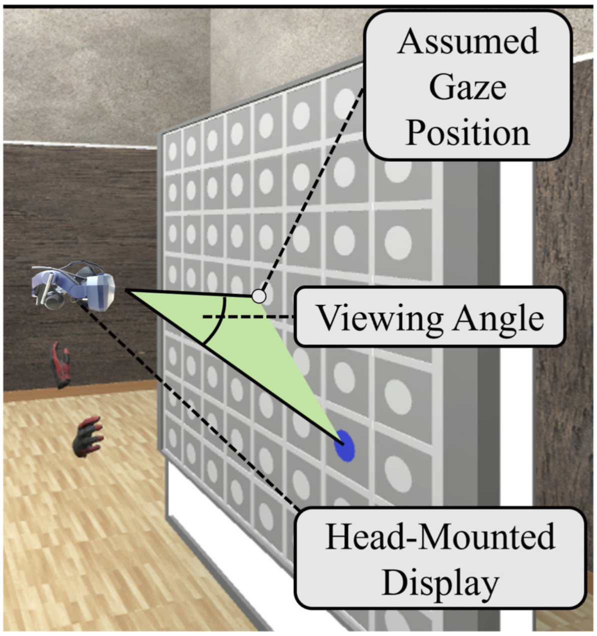

FIGURE 3
Visualization of the assumed gaze position, used because the HMD did not provide reliable eye tracking, and the viewing angle. The assumed gaze position corresponds to the straight-ahead direction and was defined as the center of the display. The viewing angle was calculated as the 3D angle between the assumed gaze position, the HMD, and the stimulus (blue circle).

Furthermore, RTs shorter than 0.15 s were considered anticipated (Collins and Long, 1996), and those longer than 1.4 s were considered missed (SCHUHFRIED GmbH, 2017) and excluded from the analysis. Outliers were eliminated using the median absolute deviation (MAD (Leys et al., 2013)). Since it was expected that RTs would increase with eccentricity (Strasburger et al., 2011), RTs were first ordered according to the corresponding viewing angle (VA). Subsequently, the MAD was applied within blocks of 100 reactions. This was done separately for each test condition. Furthermore, three participants were excluded from the analysis due to a large number of errors. Although error rate was not defined as an exclusion criterion, error rates exceeded 45% in one condition (IQR central-signal condition: 4%, IQR peripheral-signal condition: 5.56%, IQR combined: 5.06%) and could be clearly seen as outliers, different from all the others. All exclusions were made oblivious to conditional effects. To examine the robustness toward trial-exclusion, a sensitivity analysis using the dataset before and after trial-exclusion is included.

For each trial, a VA was calculated between the stimulus, the HMD, and the approximation of the gaze position. An approximation was used, as the HMD did not provide sufficient eye tracking. Therefore, the approximation of the gaze position was the point on the reaction wall, in the center of the HMD's FOV. The VA of a trial was calculated when the stimulus was first visible in the HMD. As the peripheral vision was meant to be examined in this study, reactions to stimuli with VAs smaller than 8° were excluded from the analysis. The assumed gaze position and the calculated VA are presented in Figure 3.

All RTs (with corresponding VAs) were then categorized by controller used (left, right), round (first, second, third 1-min round), and condition (no-signal, central-signal, peripheral-signal).

2.6 Statistical analysis

Data analysis was performed blinded to participant identity by using coded IDs. RTs were analyzed using a linear mixed-effects model (LMM), allowing continuous VA analysis and accounting for random variations between participants. As the RTs were not normally distributed ($p < 0.05$) and right-skewed, they were transformed using the natural logarithm, resulting in the dependent variable LnRT. Furthermore, to ensure numerical stability, the VA was z-transformed (VA_transf).

The formula of the used LMM is shown in Equation 1. The fixed effects of the LMM were built using VA_transf (continuous) and the following categorical factors: Hand (dominant, non-dominant), Round (first, second, third), Signal (no-signal, central-signal, peripheral-signal), and Test Order (individual order of the tests). Hand and Test Order did not improve the model (verified using AIC and BIC) and were therefore excluded to simplify the model. Furthermore, to investigate whether the influence of stimulus eccentricity is equal in all tests, the interaction between VA_transf and Signal was added. The categorical factors in the model were effect coded. The random effect's structure was built as follows: To account for random variation between the participants, VA_transf was implemented in the LMM as a random intercept and random slope regression. The intercept reflects the overall mean LnRT, and the slope indicates the individual increase in LnRT with eccentricity. Furthermore, to consider correlations of the effect of the Signal within participants, a random effect of Signal nested in participants was added to the model. From that, the following formula results for the R-code:

$$LnRT \sim VA_{transf} + Signal + Round + Signal{:}\, VA_{transf} + \left(1 + VA_{transf} \,\|\, Participant\right) + \left(1 \mid Participant{:}\, Signal\right) \quad (1)$$

For the fixed effects, estimates from the model were calculated, which refer to the slope of LnRT for VA_transf and to contrasts for Signal and Round. Degrees of freedom were computed using the Satterthwaite approximation. As an effect size, Cohen's f^2 was calculated using the package 'effectsize' in R (0.02 small effect, 0.15 medium effect, 0.35 large effect, (Cohen, 1988)).

Since the estimates of the fixed effects are on a log scale and thus difficult to interpret, they are transformed back from LnRT to RTs using exp (). This way, the linear effects of the estimates are translated to multiplicative effects on the original RT scale. The results achieved are, however, only valid when all other factors in the reactions are kept constant. For the sensitivity analysis, both datasets are analyzed with the same model structure.

For descriptive statistics and error rates, the reactions are categorized according to their VA into near-peripheral (8°–30°), mid-peripheral (30°–60°), and far-peripheral (beyond 60°), as proposed by Simpson (2017).

The SSQ scores before and after the VR exposure were calculated as proposed by Kennedy et al. (1993). Due to outliers, Wilcoxon signed-rank tests were used to identify significant changes in cybersickness symptoms, using the R package sjstats.

For all statistical tests, the significance level was $\alpha = 0.05$.

3 Results

3.1 Descriptive statistics

In total, 5,584 reactions were analyzed with the LMM. Of these, 2,840 were recorded in the no-signal condition (925 in Round 1, 948 in Round 2, 967 in Round 3, 393 near-peripheral, 1,558 mid-peripheral, 889 far-peripheral), 1,354 in the central-signal condition (414 in Round 1, 459 in Round 2, 481 in Round 3, 158 near-peripheral, 726 mid-peripheral, 481 far-peripheral), and 1,390 in the peripheral-signal condition (434 in Round 1, 478 in Round 2, 478 in Round 3, 211 near-peripheral, 738 mid-peripheral, 441 far-peripheral). The number of reactions in the no-signal condition is larger due to the absence of No-Go trials. The mean VA over all reactions was 49.80° (SD = 16.72°), with a mean overall RT of 0.437 s (SD = 0.134 s).

Figure 4 presents boxplots of RTs in the three test conditions (no-signal, central-signal, peripheral-signal), separated into three eccentricity regions proposed by Simpson (2017) (8°–30°: near-peripheral, 30°–60°: mid-peripheral, >60°: far-peripheral).

A scatter plot with all 5,584 reactions displaying the RTs in dependence on the VA is shown in Figure 5, together with predicted mean RT trends for each test condition.

3.2 Influence of different factors on reaction time

3.2.1 Random effects of linear mixed model analysis

Random intercepts and slopes for VA_transf were modeled per participant (Intercept: SD = 0.098, Slope for VA_transf: SD = 0.021). In addition, a random intercept for Signal nested within participant (Intercept: SD = 0.040) was included to account for repeated measures across test conditions. The standard deviation of the residuals was 0.225.

3.2.2 Fixed effects of linear mixed model analysis

The results of the fixed effects are summarized in Table 1. LnRT was significantly influenced by VA_transf and the Signal, with large effect sizes. In contrast, no statistical evidence for an influence of the factor Round and the interaction of VA_transf and Signal was observed. The estimate of VA_transf can be translated to an increase of 14% (exp (0.131) = 1.140) per standard deviation (16.72°), which equals an increase of approximately 8.1% in RT per 10° increase in eccentricity (like for all the other values, assuming all other predictors are held constant). Regarding the factor Signal, the effect coded estimates in Table 1 for No-Signal and Central-Signal can be used to calculate the estimate for Peripheral-Signal: (No-Signal + Central-Signal) = -(-0.080 + 0.008) = 0.072. These estimates represent the deviation from the overall mean and can be used to calculate the LnRT difference between the conditions and translate that to multiplicative differences on the RT scale:

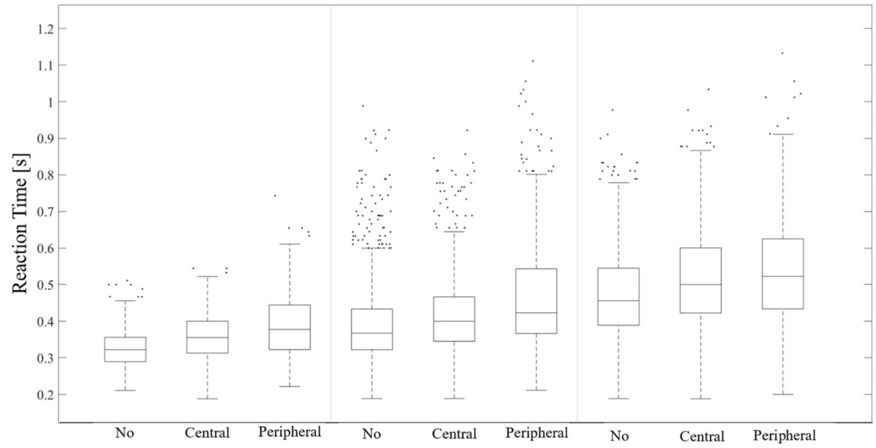

FIGURE 4
Boxplots of reaction times (go-signal or no signal at all) across the three conditions (no-signal, central-signal, peripheral-signal), separated into eccentricity regions according to Simpson (2017) (8°–30°: near-peripheral, 30°–60°: mid-peripheral, >60°: far-peripheral). This subdivision was applied solely for visualization and was not used in the statistical analyses.

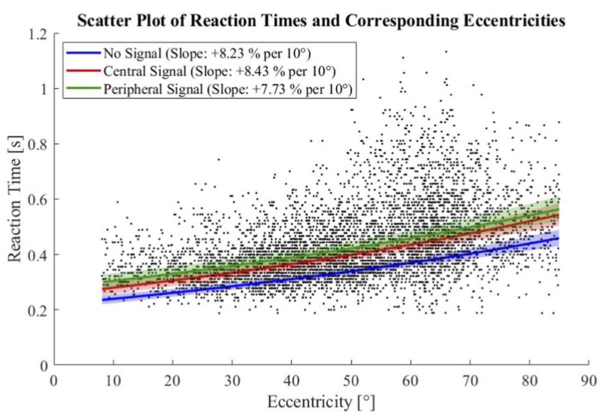

FIGURE 5
Scatter plot of all reaction times in relation to viewing eccentricity. Additionally, predicted mean reaction time trends and 95% confidence intervals as well as the displayed slopes derived from the linear mixed-effects model are overlaid, illustrating the increase in reaction times with eccentricity across the three signal conditions. Values were back-transformed from LnRT to RT using exp.

- RTs were 8.4% faster (exp (-0.080 – 0.008) = 0.916) in the no-signal condition than in the central-signal condition,
- RTs were 14.1% faster (exp (-0.080 – 0.072) = 0.859) in the no-signal condition than in the peripheral-signal condition,
- RTs were 6.2% faster (exp (0.008–0.072) = 0.938) in the central-signal condition than in the peripheral-signal condition.

The calculated f^2 based on the log-transformed RT and a large number of trials, which is why the effect sizes for VA_transf and Signal are very large as interpreted by Cohen (1988). In contrast, the very small effect size for the factor Round and the interaction between VA_transf and Signal may be based on an imprecise estimation and limited sensitivity.

The sensitivity analysis revealed that the direction and significance of the main effects (VA_transf and Signal) as well as the absence of an interaction were consistent across both datasets. However, contrary to the results presented in Table 1, a minor effect of Round (p = 0.03) was observed for the dataset without trial-exclusion.

3.3 Errors in the No-Go-conditions

Participants mistakenly reacted 22 times (error rate: 2.6%) in the central-signal condition (3 near-peripheral (3.0%), 17 mid-peripheral (3.6%), 2 far-peripheral (0.8%)) and 26 times (3.3%) in the peripheral-signal condition (2 near-peripheral (1.9%), 15 mid-peripheral (3.2%), 9 far-peripheral (4.2%)).

3.4 Cybersickness

The SSQ scores before and after the VR exposure and their statistical comparison are displayed in Table 2, showing no significant changes. As most participants reported no symptoms, median scores were typically zero. Only for one participant, the SSQ score increased by more than 15 points (defined as 'Symptoms are a concern' (Kennedy et al., 2003)), however, that participant did not complain about feeling unwell. No adverse events were reported.

4 Discussion

This study examined the influence of the location of additional Go/No-Go-signals on the RT to stimuli appearing at different eccentricities in the visual field, using a virtual reaction wall in VR. Three reaction tests were conducted: One test only involved reacting to a stimulus on the reaction wall (no-signal condition). In the other two tests, an additional signal in the central (central-signal condition) or peripheral visual field (peripheral-signal condition) indicated whether participants should react to a stimulus on the reaction wall (green additional signal) or inhibit the response (red additional signal). Due to the size of the reaction wall, the stimuli

TABLE 1 Results of the linear mixed-effects model for the dependent variable LnRT. Effect sizes were calculated on term level, which is why the f^2 refers to the complete factor (respectively interaction).

Fixed effects	Estimate [95% CI]	Std. Error	df	*t*-value	*p*	f^2
(Intercept)	−0.841 [-0.88, −0.80]	0.020	26.04	−42.835	<0.001	---
VA_transf	0.131 [0.12, 0.14]	0.005	27.88	24.886	<0.001	22.213
No signal	−0.080 [-0.09, −0.07]	0.007	45.72	−10.789	<0.001	2.638
Central signal	0.008 [-0.01, 0.02]	0.008	57.38	1.046	0.300	
Round 1	0.002 [-0.01, 0.01]	0.004	5,528	0.540	0.589	<0.001
Round 2	−0.005 [-0.01, 0.00]	0.004	5,516	−1.065	0.287	
VA_transf: No signal	0.002 [-0.01, 0.01]	0.004	5,552	0.388	0.698	<0.001
VA_transf: Central signal	0.005 [0.00, 0.01]	0.005	5,547	0.952	0.341	

TABLE 2 Median (Mdn) and median absolute deviation (MAD) from the median of the SSQ scores before and after VR exposure, along with their comparison. The z-values were approximated, and the p-values were based on asymptotic approximation. The test statistic V represents the positive rank sum of the difference between pre and post VR.

SSQ scores (n = 27)	Nausea *Mdn (MAD)*	Oculomotor *Mdn (MAD)*	Disorientation *Mdn (MAD)*	Total score *Mdn (MAD)*
Pre VR	0 (0)	0 (0)	0 (0)	3.740 (3.740)
Post VR	0 (0)	7.580 (7.580)	0 (0)	0 (0)
Wilcoxon test	$V = 12$ $z = 0.050$ $p = 0.824$	$V = 40$ $z = -1.619$ $p = 0.252$	$V = 12.5$ $z = -1.067$ $p = 0.251$	$V = 43$ $z = -1.451$ $p = 0.200$

could appear in different regions of the visual field. As the theories under investigation are based on older work and mostly rely on motorically simple responses (Danek and Mordkoff, 2011; Donders, 1969; Facoetti, 2001), VR enables including full-body movements in reactions, bringing the task closer to RW sports than simple button presses. A sensitivity analysis suggests that the main conclusions are not driven by trial-exclusion.

Regarding expectation 1), as assumed, the RTs increased with eccentricity in all test conditions, as observed in several studies before with simple (Strasburger et al., 2011) and Go/No-Go stimuli (Sun and Varshney, 2018; Thorpe et al., 2001). Similar results have been reported for a simple but sport-specific reaction test in VR (Vater, 2024) using a CAVE, particularly at large eccentricities (>60°). This eccentricity effect is typically attributed to the distribution of photoreceptor and ganglion cells in the human retina (Uemura et al., 2012; Thorpe et al., 2001) and decreased attentional abilities in the periphery (Staugaard et al., 2016).

Also, following expectation 2), RTs were shortest in the no-signal condition, followed by the central-signal condition, and were slowest in the peripheral-signal condition. This confirms findings from previous work, showing longer RTs in Go/No-Go tasks compared to simple reaction tasks (Squella and Ribeiro-do-Valle, 2003). The increased RTs in Go/No-Go tasks are thought to result from stimulus-signal processing delays associated with the need to process inhibitory signals. Only once a stimulus is clearly identified as a go-signal can the motor response be released (Danek and Mordkoff, 2011), which is consistent with our data.

The observed RT difference between the two additional signal conditions (RTs in central-signal condition being 6.2% faster than in peripheral-signal condition) was also expected (expectation 3). A possible reason could be that the peripheral signal required more time to be detected, following the same logic as the eccentricity effect. The delay in discriminating the peripheral signal could explain the longer RTs in this condition. This interpretation is supported by several studies showing that stimuli presented in peripheral vision lead to slower responses than foveal stimuli (Uemura et al., 2012). It is difficult to determine whether the difference of 6.2%, which is only valid under the assumptions of the LMM, is meaningful in sports. Considering an RT of 0.4 s, an increase of 6.2% would correspond to 24.8 ms. For some sports, especially martial arts, this time interval may be meaningful. If, for example, a punch in boxing has a velocity of approximately 7 m/s (Kimm and Thiel, 2015), the fist travels approximately 17 cm in that time. Of course, that is only a theoretical assumption, as further processes influence whether an athlete gets hit or not.

Another factor explaining the difference in RT between the central-signal and peripheral-signal conditions may be the reduced color discriminability in the peripheral visual field (Hansen et al., 2009). Not only is the detection of peripheral stimuli slower, but color discrimination may be less efficient, delaying the decision to respond or inhibit. This is also reflected by a slightly higher error rate in the peripheral-signal condition compared to the central-signal condition. Nevertheless, red and green are still reliably discriminable up to eccentricities of 50° (Hansen et al., 2009), which aligns with our low error rates.

The explorational examination, whether the additional signals shift attention and facilitate or limit responses in the respective areas, did not provide statistical evidence for an influence of the additional signal's position. This differs from what has been observed for task-irrelevant signals, which influenced RTs (Squella and Ribeiro-do-Valle, 2003). However, possibly the used test and measured data may not be sensitive enough to show such a connection. A further possible explanation is that in our setup, stimuli could appear anywhere across the wall and not only near the location of the additional signal, potentially diffusing attentional focus. Additionally, the error rates demonstrate the opposite of what we would have expected, as they increase with eccentricity in the peripheral-signal condition but not in the central-signal condition. However, due to the low total number of errors, the distribution over the visual field should be interpreted with caution.

Another reason why RTs are not observed to be facilitated by the additional peripheral signals in the peripheral visual field might be the bilateral presentation of the additional signal. Different outcomes might be observed with the unilateral presentation of additional signals, possibly facilitating reactions on the side of the visual field where this signal is shown. Furthermore, in sports, the additional signal, indicating to react or withhold a response, is mostly not represented by a sphere. As different stimuli shapes, sizes, contrasts, and movement speeds may influence RTs (Plainis and Murray, 2000; Strasburger et al., 2011), this is possibly also the case for additional Go/No-Go-signals. In addition, a more controlled test design excluding head movements and showing stimuli and cues at specific locations could better test for such an effect.

Contrary to the results of many reaction tests (Schumacher et al., 2019), no learning effect was observed, neither within a test (no observable effect of Round) nor across tests (test order did not improve LMM). However, the sensitivity analysis using the dataset without any trial removal yields a significant effect of Round, indicating that the data of the final analysis possibly insufficiently depict this effect. The inclusion of practice trials could therefore be reasonable in future studies.

In the feedback questionnaire after each condition, the participants rated the respective condition from 1 (very easy) to 5 (very difficult). In line with the differences in RTs, the no-signal condition was rated as the easiest ($M = 1.9$, $SD = 0.7$), followed by the central-signal condition ($M = 2.1$, $SD = 0.6$), and the peripheral-signal condition ($M = 2.4$, $SD = 0.9$). Interestingly, some participants reported that the central additional signal was more distracting, while the peripheral additional signal led attention toward the peripheral stimuli and therefore facilitated the search there. However, these subjective impressions were not reflected in the RT or error data.

5 Limitations and future directions

Some limitations must be acknowledged. First, the HMD used in this study lacks sufficient eye tracking, which is why only an assumed position of the gaze is used for VA calculation. The HMD was chosen, as it provides the greatest known FOV. However, the expected results show patterns consistent with RW findings, as the VA had a strong influence on the RT, and the results were consistent across the three different conditions. Moreover, it has been observed that eye movements are often supported by head movements, in order to keep the eye in a neutral position (Stahl, 1999) in the central 25° eccentricity (Sidenmark and Gellersen, 2020). However, in search tasks, head movements are often preceded by eye movements (Freedman, 2008), which is why adding eye tracking to the setup could be interesting.

Second, unknown and possibly inconsistent latencies of the application could have influenced the RT measurements, although efforts were made to minimize this factor by efficient modelling and programming. Here again, the consistent results across the three conditions and the absence of an effect of round indicate a negligible influence of latencies. Possibly, when the reaction tests are used to monitor RTs over several sessions, latencies may become more problematic and should be addressed. Further technical restrictions, like display resolution and reduced FOV compared to the RW, could limit the transferability of results into the RW.

Third, integrating practice trials could have been advantageous. Although no learning effect was found in our analysis (factor Round), the sensitivity analysis using the dataset without any exclusions showed shorter RTs over rounds. Therefore, the integration of practice trials could have led to more stable results and prevented the high error rates of the excluded participants.

Fourth, the used sample mainly consisted of males with VR experience, limiting generalizability.

Fifth, re-centering the head's position at each trial start could have led to more standardized results, thus limiting the dynamic test procedure.

Future studies should explore more sport-specific applications of this paradigm. For example, in a basketball defense scenario, a participant could focus on the ball carrier while responding to other players cutting toward the basket, only reacting when no teammate is already covering that action. A VR-based system could be developed as a training tool for improving decision-making in dynamic sports contexts, without requiring the physical presence of teammates or opponents.

6 Conclusion

The study demonstrates that RTs increase with stimulus eccentricity and that RTs are longer in Go/No-Go-like tasks compared to simple reaction tasks on a VR reaction wall, mirroring results from RW settings. Furthermore, no evidence was found for an influence of the position of an additional signal, indicating whether to react or not, on the slope of the eccentricity-related increase of RT, suggesting no support for attentional facilitation by the additional signal under the present conditions.

The observed results were measured in VR, but sport takes place in RW. Since many studies before reported correlations between RTs in VR and RW (Bürger et al., 2024; Langer et al., 2023; Pastel et al., 2025a), the observed results suggest potential transferability, especially given the generally low levels of cybersickness and confirming patterns from RW studies like increasing RTs with increased stimulus eccentricity (Strasburger et al., 2011) and longer RTs in Go/No-Go-like tasks compared to simple reactions (Squella and Ribeiro-do-Valle, 2003). This supports the usage of VR in tests and training of cognitive functions.

Data availability statement

The raw data supporting the conclusions of this article will be made available by the authors, without undue reservation.

Ethics statement

The studies involving humans were approved by Ethik-Kommission der Otto-von-Guericke-Universität an der Medizinischen Fakultät und am Universitätsklinikum Magdeburg A. ö. R. [Ethics Committee of the Otto von Guericke University at the Faculty of Medicine and University Hospital Magdeburg (public corporation)]. The studies were conducted in accordance with the local legislation and institutional requirements. The participants provided their written informed consent to participate in this study.

Author contributions

DB: Conceptualization, Data curation, Formal Analysis, Investigation, Methodology, Software, Validation, Visualization, Writing – original draft. KA: Investigation, Writing – review and editing. FH: Conceptualization, Methodology, Writing – review and editing. SP: Conceptualization, Methodology, Validation, Writing – review and editing. KW: Conceptualization, Funding acquisition, Methodology, Project administration, Supervision, Writing – review and editing.

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Conflict of interest

The author(s) declared that this work was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

Generative AI statement

The author(s) declared that generative AI was not used in the creation of this manuscript.

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Supplementary material

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